

Data Structures & Algorithms

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1. Each node of a Singly Linked List contains:
⇒ ~~(b)~~ Data and a pointer to the next node.
2. The next field of the last node in a singly linked list ~~stores~~ is:
⇒ ~~(b)~~ NULL
3. A structure that contains a pointer to a variable of its own type is called.
⇒ Self-referential structure
4. A Singly Linked List is empty when:
⇒ $head == NULL$
5. Which of the following correctly inserts newnode at the beginning of a list with head pointer head?
⇒ $newnode \rightarrow next = head; head = newnode;$
6. Which statement sequence correctly deletes the first node of a non-empty list?
= $temp = head; head = head \rightarrow next; free(temp);$
7. Which operation is not directly possible in a singly linked list?
= Traversing from tail to head
8. In a singly linked list of n nodes, the total number of next pointer fields is:
⇒ n
9. Given a pointer p to a node that is not the last node, and ~~no~~ access to the head pointer, the node can be deleted by:
⇒ Copying the data of $p \rightarrow next$ into p , then deleting $p \rightarrow next$.
10. The connect loop condition to traverse and print all nodes using a pointer p initialised to head is:
⇒ $while (p \neq NULL)$
11. To insert a node at the end of a singly linked list, traversal must continue until reaching the node where:
= $p \rightarrow next == NULL$
12. Floyd's cycle detection algorithm works by using:
= Two pointers advancing at different rates.

13. Iterative reversal of a singly linked list typically requires how many auxiliary pointers?

= Three

14. The condition $\text{head} \neq \text{NULL}$ & $\text{head} \rightarrow \text{next} = \text{NULL}$ indicates that the list has:

→ Exactly one node

15. Which C statement correctly allocates memory for one node?

⇒ $p = (\text{struct node} *) \text{malloc}(\text{sizeof}(\text{struct node}));$

16. The main purpose of using a dummy (header) node is to:

→ Avoid special case handling for insertion and deletion at the head.

17. In a circular singly linked list, the next field of the last node points to:

= The first node

18. Which of the following is a genuine disadvantage of a singly linked list compared to an array?

⇒ It does not support random (indexed) access

19. If the head pointer of a singly linked list is accidentally overwritten without saving it, the result is:

= Loss of access to all nodes, causing a memory leak

20. To find the middle node of a singly traversal, one may use:

⇒ A slow pointer and a fast pointer.

21. After executing $p = \text{head}; \text{while}(p \rightarrow \text{next} \neq \text{NULL}) p = p \rightarrow \text{next};$ on a non-empty list the pointer p refers to:

= The last node

22. Which of the following statements is true about a singly linked list?

= Nodes may be scattered anywhere in memory.

23. Write the C structure declaration for a node of a singly linked list stores an integer

```
struct Node {  
    int data;  
    struct Node * next;  
};
```

24. Consider the following code fragment. What is the output?

List 10 → 20 → 30 → 40 → NULL

The loop prints:

10 + 20 = 30

20 + 30 = 50

30 + 40 = 70

output 30 50 70

25. Struct node *insertEnd (struct node *head, int x) {
 Struct node *t = (struct node *) malloc (sizeof(struct node));
 t->data = x;
 t->next = NULL;
 if (head == NULL) {
 return t;
 }
 Struct Node *p = head;
 while (p->next != NULL) {
 p = p->next;
 }
 p->next = t;
 return head;
}

26. Struct Node *reverseList (Struct Node *head) {
 Struct Node *prev = NULL, *current = head, *next = NULL;
 while (current != NULL) {
 next = current->next;
 current->next = prev;
 prev = current;
 current = next;
 }
 return prev;
}

27.

27. Explain, with

A → B → C → NULL

To delete B, change the pointers of A:

Before: A → B → C → NULL

After: A → C → NULL

B → Deleted

Pointer adjustment

A → next = B → next;

free(B);

28. A singly linked list contains 5 → 8 → 12 → 3 → NULL

(i) Insert 7 at the beginning 7 → 5 → 8 → 12 → 3 → NULL

(ii) Delete the node containing 12: 7 → 5 → 8 → 3 → NULL

(iii) Insert 9 at the end: 7 → 5 → 8 → 3 → 9 → NULL